

Basic Electrical Engineering

DC Circuit

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Resistance:- it is the property of material by virtue of which it opposes the flow of current.

$$R = \frac{\rho l}{a}$$

$$R_t = R_0 (1 + \alpha (T_2 - T_1))$$

OHM's Circuit:- A source which is connected with load resistance.

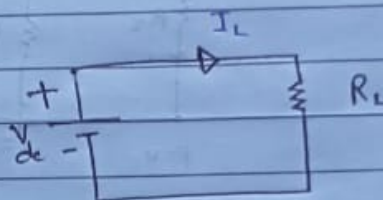
OHM's Law:-

The current flowing through the conductor is directly proportional to p.d across the load in a loop with all the physical parameter remains constant.

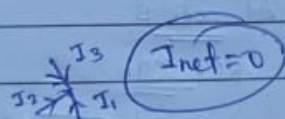
$$I \propto V$$

$$R \propto V/I$$

$$R \propto V/I$$



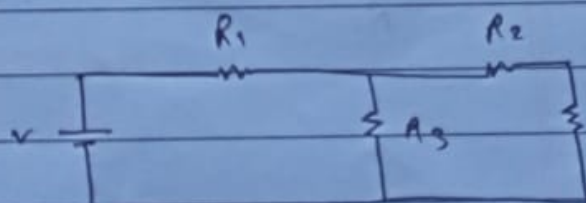
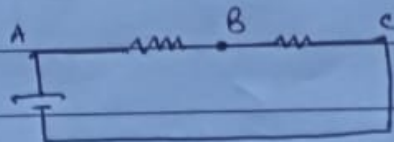
KCL:-



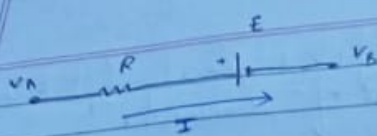
→ Junction:-

at least 2 branch need.

Node:- at least 3 branch need and have potential.



in ideal case, at a junction the algebraic sum of all the current will be zero.

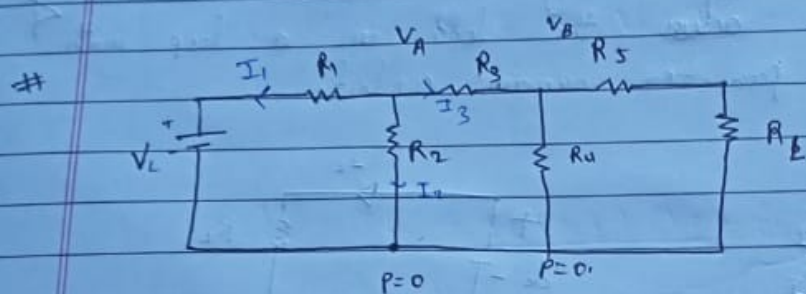


$$V_A - (IR) - (E) - V_B = 0$$

$$V_A - V_B = E + IR$$

a KVL in any closed circuit or mesh the algebraic sum of all the $\downarrow$ Emfs & voltage drop $\downarrow$ will be zero.

$$\sum \text{Emfs} + \sum IR = 0$$



At Node A

KCL Eqn

$$I_1 + I_2 + I_3 = 0$$

Ohm Law :

$$V = IR \quad I = V/R$$

$$\therefore \frac{V_A - V_L}{R_1} + \frac{V_A}{R_2} + \frac{V_A - V_B}{R_3} = 0$$

At node B

KCL Eqn

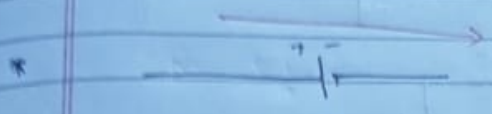
$$I_3 + I_4 + I_5 = 0$$

$$I_3 + I_4 + I_5 = 0$$

$$\therefore \frac{V_B - V_A}{R_3} + \frac{V_B}{R_4} + \frac{V_B}{R_5 R_L} = 0$$



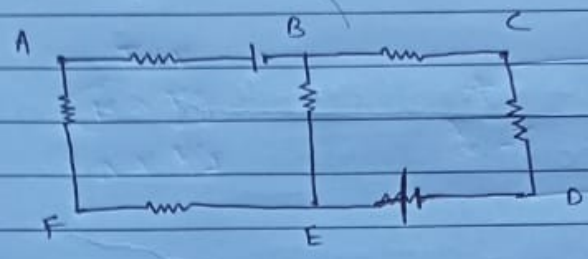
Rise in potential (+ve)



Drop in potential (-ve)

Mesh analysis method:-

Mesh: it is a closed path of Electrical circuit through which current can flow, but there will be no any other closed path inside it or across it.



Mesh

ABEFA ✓

BCDEB ✓

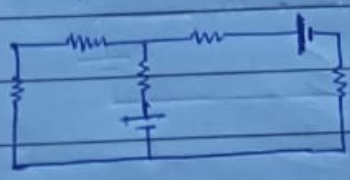
ACDFA X

→ loop

But Loop can contain various loop inside it

Mesh analysis method:-

Type-1
voltage source & resistor

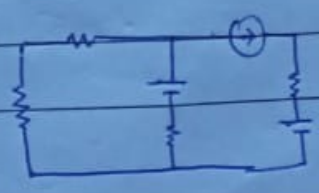


Type-2

V, R, I

it should be in outer loop:-

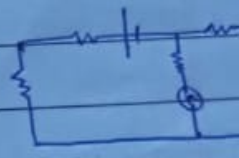
that particular branch which can't be shared by 2 loops.



Type-3

V, R, I

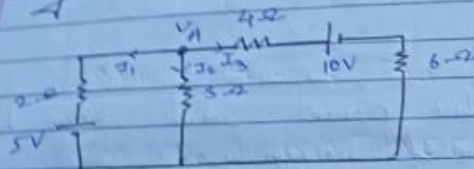
in inner loop



Nodal Method

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if using nodal def. the branch current.



A) V_A $I_1 + I_2 + I_3 = 0$

$$\frac{V_A - 5}{2} + \frac{V_A - 0}{3} + \frac{V_A - 10}{10} = 0$$

$$30V_A - 150 + 20V_A + 6V_A - 60 = 0$$

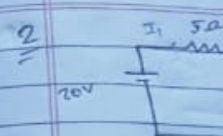
$$56V_A - 210 = 0$$

$$V_A = \frac{210}{56} = \frac{105}{28} \text{ V}$$

$$I_1 = \frac{\frac{105}{28} - 5}{2} = \frac{105 - 140}{28} \times \frac{1}{2} = \frac{-35}{56} = -\frac{5}{8}$$

$$I_2 = \frac{\frac{105}{28} \times \frac{1}{3}}{\frac{1}{3}} = \frac{5}{4}$$

$$I_3 = \frac{105 - 280}{10 \times 28} = \frac{-175}{280} = -\frac{5}{8}$$



KCL At node $I_1 + I_2 + I_3 = 0$
 $V_A - 20 = 0$

$$10V_A = 200$$

$$V_A = 20$$

$$\Rightarrow$$

$$\Rightarrow$$

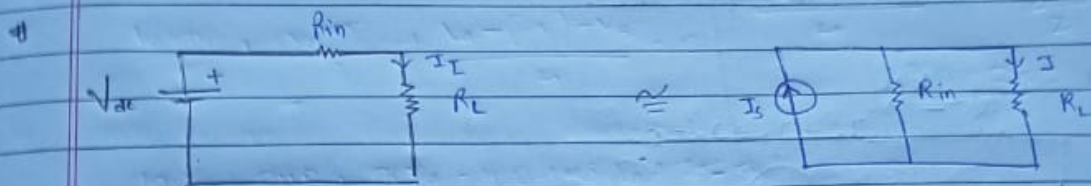
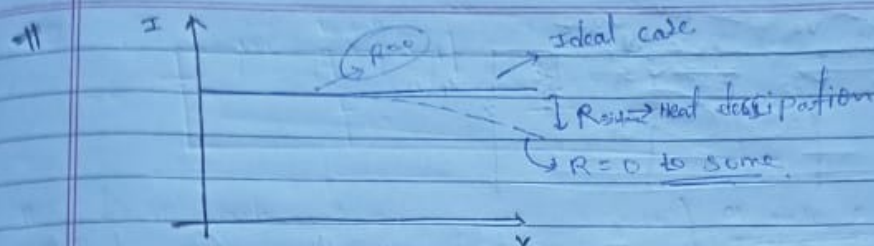
$$\Rightarrow B$$

$$49V_A - 15 \left(\frac{60 + 2V_A}{7} \right) = 265$$

$$329V_A - 900 + 30V_A = 1855$$

$$359V_A = 1855 + 900$$

$$V_A = \frac{2755}{359}$$



$$I_L = \frac{V_{dc}}{R_{in} + R_L}$$

using current division Rule

$$I_L = \left(\frac{R_{in}}{R_{in} + R_L} \right) I_s$$

compare

$$V_{dc} = I_s R_{in}$$

$$I_s = \frac{V_{dc}}{R_{in}}$$

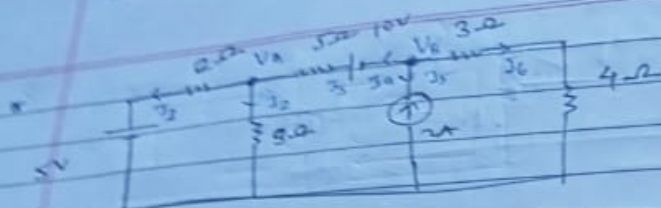
There should be no any other Resistance bet
current source & int. Resistance,
else,
we can't able to convert it in voltage
source.

can't count
P.D.

$$I = \frac{V}{R}$$



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using the Nodal analysis determine the Current across 4Ω Resistance.

using KCL

$$I_1 + I_2 + I_3 = 0$$

$$\frac{V_A - 5}{2} + \frac{V_A}{3} + \frac{V_A - 10 - V_B}{5} = 0$$

$$\frac{15V_A - 75 + 10V_A + 6V_A - 60 - 6V_B}{30} = 0$$

$$\Rightarrow 31V_A - 6V_B - 135 = 0$$

$$\Rightarrow 31V_A - 6V_B = 135$$

$$I_4 + I_5 + I_6 = 0$$

$$\frac{V_B + 10 - V_A}{5} + (-2) + \frac{V_B}{7} = 0$$

$$\frac{7V_B + 70 - 7V_A - 70 + 5V_B}{35} = 0$$

$$12V_B - 7V_A = 0$$

$$V_B = \frac{7V_A}{12}$$

$$= \frac{7 \times 34}{12} = \frac{63}{22}$$

Current

$$3V_A - \cancel{4} \times 7V_A = 1.85$$

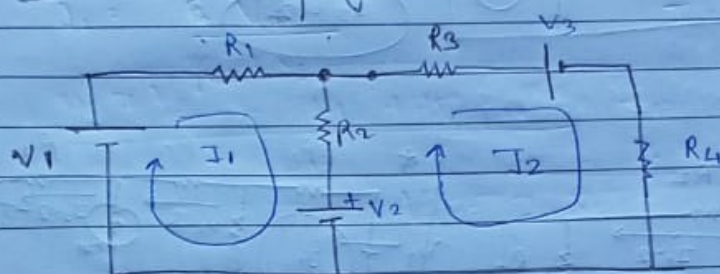
$$\frac{62V_A - 7V_A}{2} = 1.85$$

$$55V_A = 270$$

$$V_A = \frac{270}{55} = \frac{54}{11}$$

• Node

Mesh / Loop Analysis



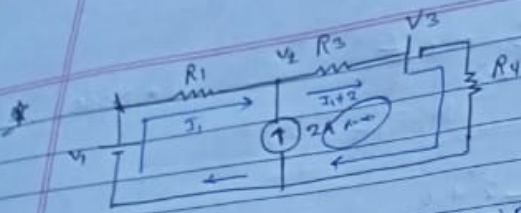
KVL (LOOP 1)

$$I_1 R_1 + (I_1 - I_2) R_2 + V_2 - V_1 = 0$$

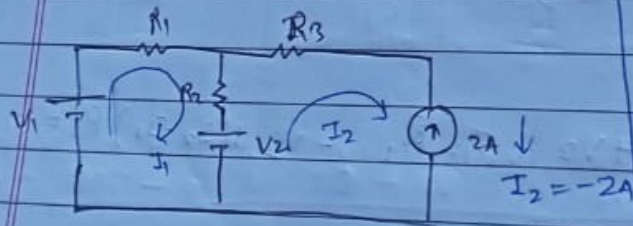
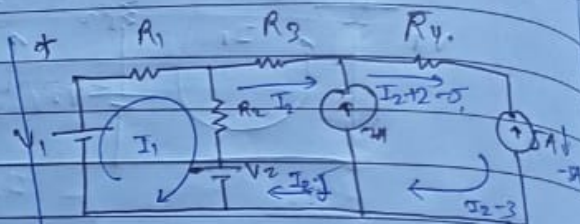
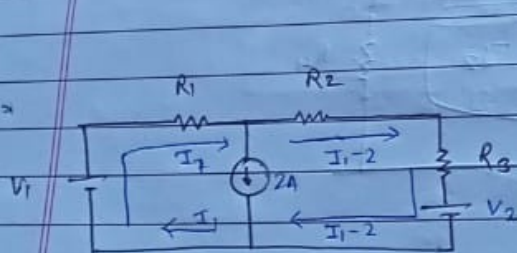
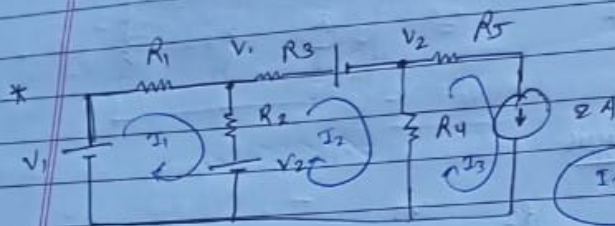
KVL (LOOP 2)

$$I_2 R_2 + V_3 + I_2 R_4 - V_2 + (I_2 - I_1) R_2 = 0$$

$$= \frac{63}{22}$$



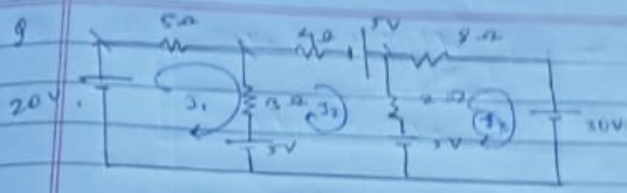
$$I_1 R_1 + (I_1 + 2) R_3 + V_3 + (I_1 + 2) R_4 - V_1 = 0$$



$$I_2 = \frac{-70 + 25}{6} = -\frac{45}{6} = -7.5$$

$$I_2 = \frac{-70 + 150}{12} = \frac{80}{12} = \frac{20}{3}$$

$$\frac{25}{16} \times 3 = \frac{75}{16}$$



using mesh analysis & loop analysis determine individual current in the loop.

using KVL Eqn for Loop 1

$$5I_1 + (I_1 - I_2)3 + 5 - 20 = 0$$

$$8I_1 - 3I_2 - 15 = 0$$

$$8I_1 - 3I_2 = 15$$

$$8I_1 - 3(10I_2 + 25) = 15$$

$$16I_1 - 30I_2 - 75 = 30$$

Loop 2

$$4I_2 - 5 + (I_2 - I_1)2 + 5 - 5 + (I_2 - I_3)2 = 0$$

$$-I_1 + 5I_2 + 9I_3 - 3I_1 = 0$$

$$-2I_1 - 3I_3 + 5I_2 = 0$$

Loop 3

$$8I_3 + 30 - 5 + (I_2 - I_3)2 = 0$$

$$10I_2 + 25 - 2I_3 = 0$$

$$I_2 = \frac{10I_3 + 25}{2}$$

$$2(10I_2 + 25) - 3I_1 + 5I_2 = 0$$

$$15I_2 + 25 - 3I_1 = 0$$

$$I_1 = \frac{15I_2 + 25}{3}$$

$$I = \frac{15 \times 7 + 25}{3}$$

$$16 \left(\frac{15I_2 + 25}{3} \right) - 30I_2 - 75 = 0$$

$$240I_2 + 400 - 30I_2 - 225 = 0$$

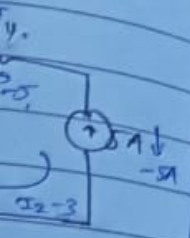
$$150I_2 + 175 = 0$$

$$I_2 = \frac{-175}{150} = -\frac{7}{6}$$

$$\frac{-25 + 50}{6}$$

$$= \frac{25}{6}$$

$$= \frac{5}{2}$$



Handwritten calculations on the left margin:

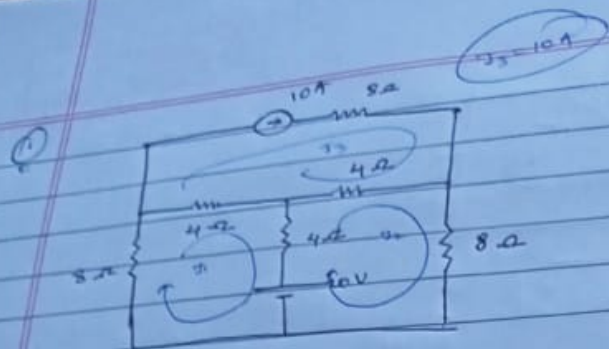
$$\frac{25}{3}$$

$$\frac{16}{150}$$

$$\frac{25}{400}$$

$$\frac{80}{150}$$

$$\frac{10}{150}$$



KVL in ①

$$4I_1 + 4(I_1 - I_2) + 20 + 8I_1 = 0$$

$$16I_1 - 4I_2 + 20 = 0$$

$$8I_1 - 2I_2 + 10 = 0$$

$$I_1 = \frac{2I_2 + 10}{8} = \frac{4I_2 - 15}{8}$$

KVL in ②

$$4I_2 + 4I_1 = 60$$

$$4I_2 - I_1 = 15$$

$$I_1 = 4I_2 - 15$$

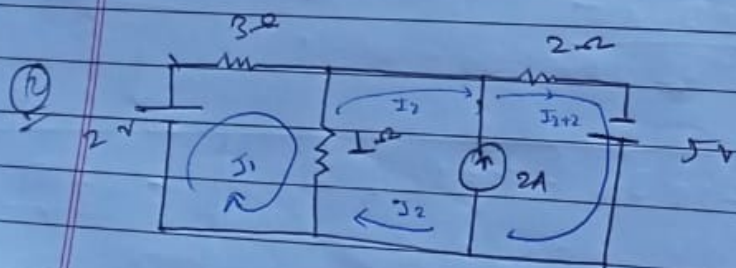
$$2I_2 + 10 = 3I_2 - 120$$

$$130 = 3I_2$$

$$I_2 = 130/3$$

$$I_1 = \frac{4 \times 130}{3} - 15$$

$$= \frac{520 - 45}{3} = \frac{475}{3}$$



$$3I_1 + I_1 - I_2 - 2 = 0$$

$$4I_1 - I_2 = 2$$

$$I_2 = 4I_1 - 2$$

$$= \frac{28 - 22}{11} = \frac{6}{11}$$

$$(I_2 + 2)(2) - 5 + I_2 - I_1 = 0$$

$$3I_2 + 4 - 5 - I_1 = 0$$

$$12I_1 - 6 - 1 - I_1 = 0$$

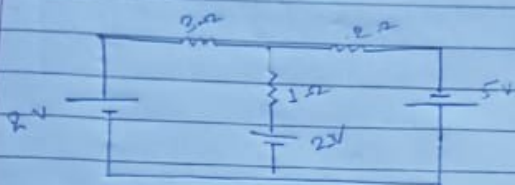
$$11I_1 = 7$$

$$I_1 = 7/11$$

across 2Ω

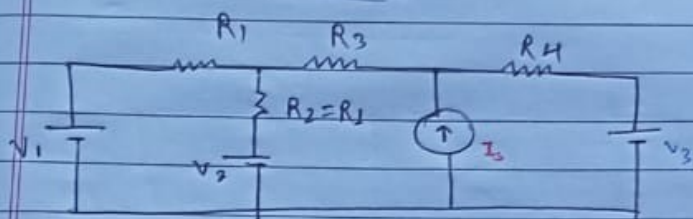
$$I_2 = \frac{28}{11}$$

using Mesh analysis:-

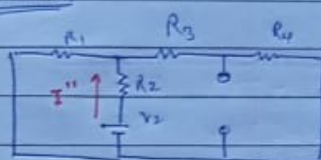


* Superposition theorem:-

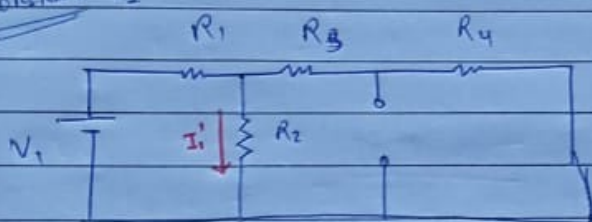
At least two sources are



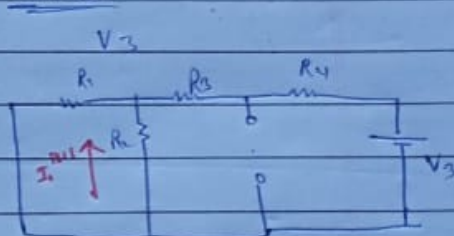
consider V_3



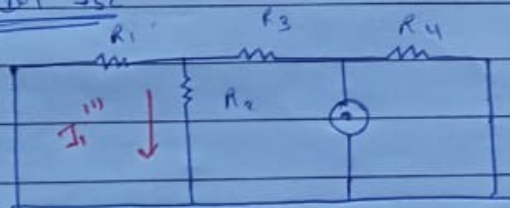
consider V_2



consider



consider I_{sc}



using Nodal, Mesh, KVL, KCL

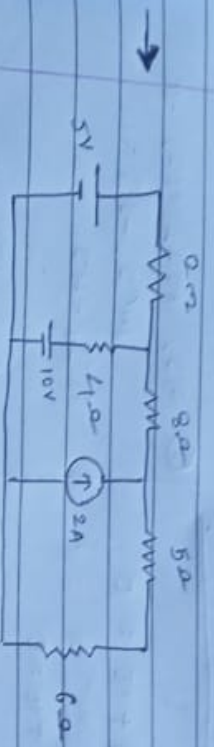
$$\Rightarrow I_1 \downarrow + I_2 \uparrow + I_3 \downarrow + I_4 \uparrow = I_{R_2}$$

$$\Rightarrow I_1 \downarrow - (I_2 \downarrow) + I_3 \downarrow - (I_4 \downarrow) = I_{R_2}$$

→ Superposition theorem:-

is at least two sources.

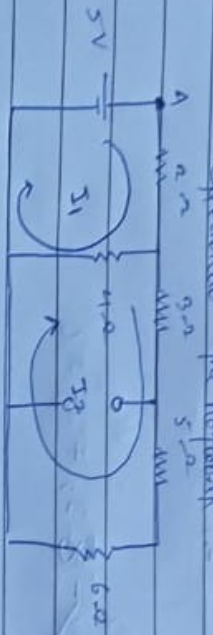
ii)



Reference the load current through 6 ohm

consider 5V

Redraw the network



KVL

$$2I_1 + 4(I_1 - I_2) - 5 = 0$$

$$6I_1 - 4I_2 = 5$$

2nd loop

$$3I_2 + 5I_2 + 6I_2 + 4(I_2 - I_1) = 0$$

$$18I_2 - 4I_1 = 0$$

$$I_1 = \frac{18}{4} I_2 = \frac{9}{2} I_2$$

$$27I_2 - 4I_2 = 5$$

$$23I_2 = 5$$

$$I_2 = \frac{5}{23} \text{ A} \downarrow$$

consider 10V



$$\textcircled{1} \quad 2I_1 + 4(I_1 - I_2) + 10 = 0$$

$$6I_1 - 4I_2 + 10 = 0$$

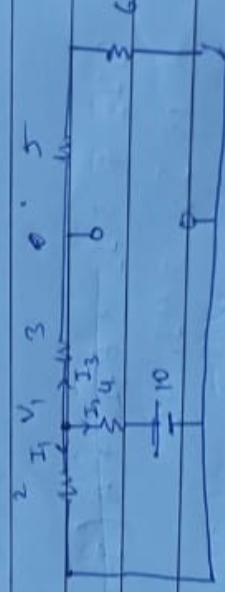
$$\textcircled{2} \quad (18I_2 - 10) - 4I_2 + 10 = 0$$

$$54I_2 - 30 - 8I_2 + 20 = 0$$

$$46I_2 - 10 = 0$$

$$I_2 = \frac{10 - 5}{23} = \frac{5}{23}$$

By Nodal



$$I_1 + I_2 + I_3 = 0$$

lim's law

$$\frac{V_1}{2} + \frac{V_1 - 10}{4} + \frac{V_1}{14} = 0$$

$$V_1 + \frac{V_1 - 10}{2} + \frac{V_1}{7} = 0$$

current solus $\rightarrow \infty$ Resistance $\rightarrow$ nodal analysis

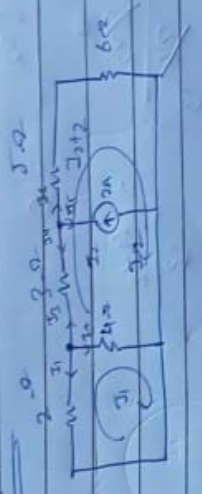
$$11V_1 + 7V_1 - 70 + 2V_1 = 0$$

$$22V_1 = 70$$

$$V_1 = \frac{70}{22}$$

$$I_3 = \frac{70 - 105}{22 \times 10} = \frac{5}{22}$$

consider 2A



$$2I_1 + 4(I_1 - I_2) = 0$$

$$6I_1 - 4I_2 = 0$$

$$3I_1 = 2I_2 \quad I_1 = \frac{2I_2}{3}$$

$$3I_2 + 5(I_2 + 2) + 6(I_2 + 2) + 4(I_2 - I_1) = 0$$

$$18I_2 + 22 - 4I_1 = 0$$

$$18I_2 + 22 - \frac{8I_2}{3} = 0$$

$$54I_2 + 66 - 8I_2 = 0$$

$$46I_2 + 66 = 0 \quad I_2 = \frac{-66}{46} = \frac{-33}{23}$$

$$I_2 + 2 = \frac{-33 + 46}{23}$$



$$= \frac{13}{23}$$

By nodal

for v_1

$$I_1 + I_2 + I_3 = 0$$

$$\frac{v_1}{2} + \frac{v_1}{4} + \frac{v_2 - v_1}{3} = 0$$

$$\frac{12v_1 + 6v_1 + 8v_1 - 8v_2}{24} = 0$$

$$\frac{26v_1 - 8v_2}{24} = 0$$

$$v_1 = \frac{8v_2}{26} = \frac{4v_2}{13}$$

At v_2

$$I_4 + I_5 + I_6 = 0$$

$$\frac{v_2 - v_1}{3} = 2 + \frac{v_2}{11} = 0$$

$$\frac{11v_2 - 11v_1 - 66 + 3v_2}{33} = 0$$

$$14v_2 - 11v_1 - 66 = 0$$

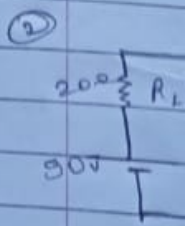
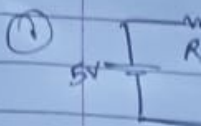
$$14v_2 - 11 \times \frac{4v_2}{13} - 66 = 0$$

$$182v_2 - 44v_2 - 858 = 0$$

$$138v_2 = 858$$

$$v_2 = \frac{858}{138} = \frac{143}{23}$$

→ Def



$$\begin{array}{r} 66 \\ 13 \\ \hline 135 \\ 66 \\ \hline 858 \end{array}$$

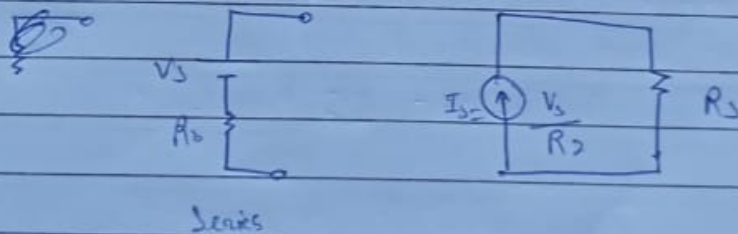
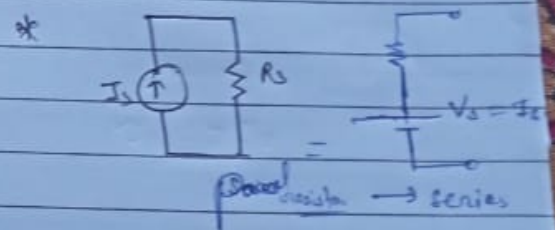
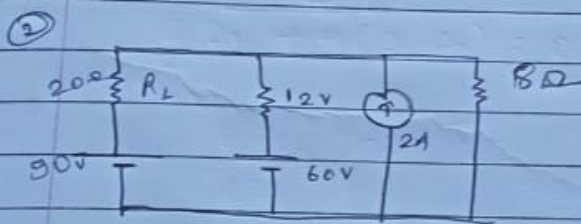
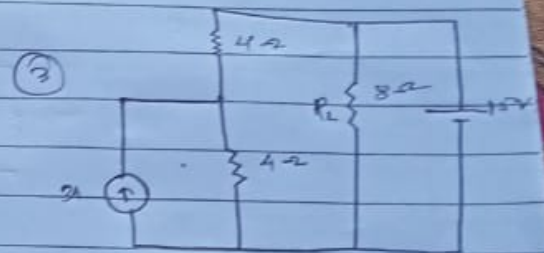
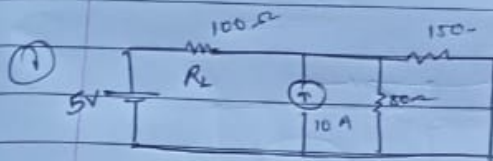
$$\begin{array}{r} 4 \\ 13 \\ \hline 47 \\ 44 \\ \hline 182 \end{array}$$

$$I_2 = \frac{V_2}{11} = \frac{143}{23 \times 11} = \frac{13}{23} \downarrow$$

$$I_T = \frac{5}{23} \downarrow + \frac{5}{23} \downarrow + \frac{13}{23} \downarrow$$

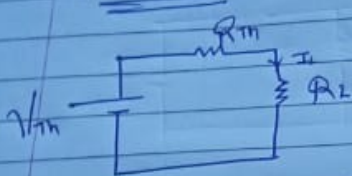
$$I_T = 1 \downarrow$$

→ Determine the load current using Superposition Theorem.



- Thermin's theorem:- Any complex circuit converted into ohm's circuit.

Ohm's circuit



$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

$R_{th} \rightarrow R_{eq}$ except R_L .

→ Active source:- it has EMF source.

Passive source:- it has no EMF source only complex resistance.

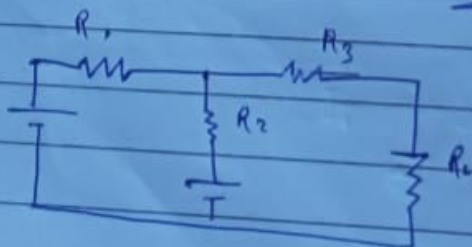
- $R_{th} \rightarrow$ Passive circuit
Given Nb

$V_{source} \rightarrow$ S.C

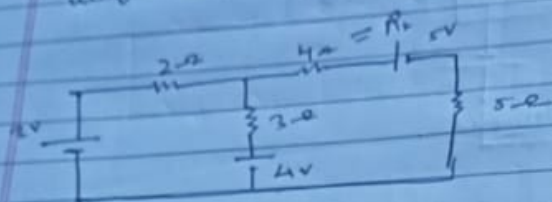
$I_{source} \rightarrow$ O.C,

Now circuit convert into Passive circuit.

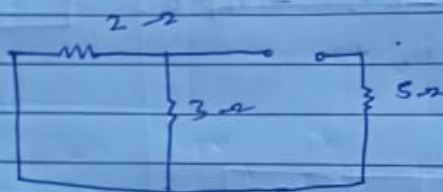
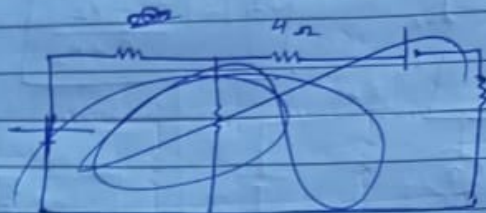
→ using series-parallel combination determine R_{th} , $R_L \rightarrow$ Removed.



- Determine the current through the load resistor using Th. I_L .



$R_{TH} = \text{resistor} = 5\Omega$

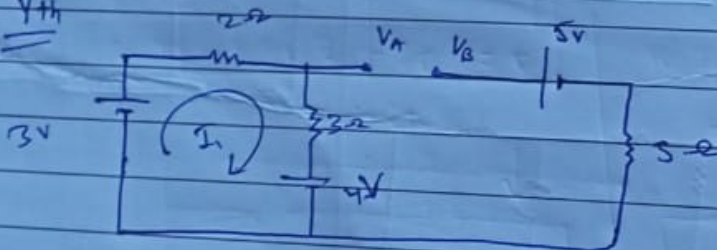


$(2 \parallel 3) + 5 = R_{TH}$

$\frac{6}{5} + 5 = R_{TH}$

$R_{TH} = \frac{31}{5}$

V_{TH}



$2I_1 + 3I_1 + 4V - 3V = 0$

$5I_1 + 1V = 0$

$I_1 = -1/5 A \downarrow$

resistance

$$V_A - \left(\frac{1}{5} \times 3\right) - 4 - \left(-\frac{5}{5}\right) = 0$$

$$V_A + \frac{3}{5} - 4 + 5 - V_B = 0$$

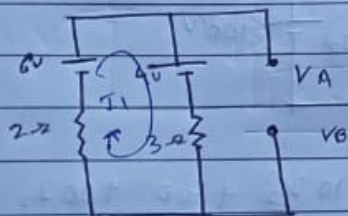
$$\Rightarrow V_A - V_B = -\frac{8}{5} \text{ V}$$

$$V_A - \left(-\left(\frac{1}{5}\right) \times 2\right) - 3 - (-5) - V_B = 0$$

$$V_A - \frac{2}{5} + 2 = V_B = 0$$

$$\boxed{V_A - V_B = -\frac{8}{5} \text{ V}}$$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{+\frac{8}{5}}{\frac{31}{5} + 4} = \frac{+\frac{8}{5}}{\frac{51}{5}} = +\frac{8}{51} \text{ A}$$



$$R_{Th} = \frac{6}{5}$$

$$\underline{I_1 = \frac{4}{5}}$$

$$V_A - 4 - \frac{2}{5} \times 3 - V_B = 0$$

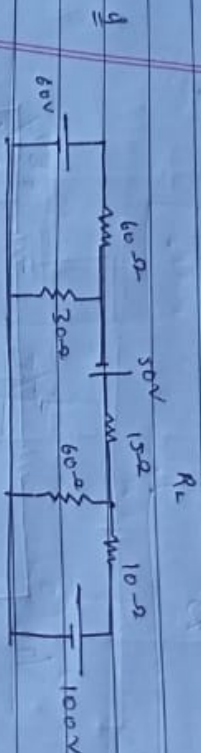
$$\boxed{V_A - V_B = \frac{26}{5}}$$

Outer loop

$$V_A - 6 \left(-\frac{2}{5} \times 2 \right) - V_B = 0$$

$$V_A - V_B = -\frac{24}{5} = -4.8$$

$$I_L = \frac{V_{Th}}{R_{Th} + R_L} = \frac{-\frac{24}{5}}{\frac{6}{5} + 4} = -\frac{24}{26} = -\frac{12}{13}$$



→ $\oint_{\text{loop}} V_{Th}$



$$60 I_1 + 30 I_1 - 60 = 0$$

$$90 I_1 = 60$$

$$I_1 = \frac{2}{3}$$

$$10 I_2 + 100 + 60 I_2 = 0$$

$$70 I_2 = -100$$

$$I_2 = -\frac{10}{7}$$

$$V_A - (50) - \left(\frac{2}{3} \times 20 \right) - \left(-\frac{10}{7} \times 60 \right) - V_B = 0$$

$$V_A - V_B = 70 - 60 + \frac{600}{7} = \frac{490 - 600}{7} = -\frac{110}{7}$$

$$V_A - V_B = \frac{70 - 600}{7} = \frac{490 - 600}{7} = -\frac{110}{7}$$

Min always for Max

~~$\frac{1}{6} + \frac{1}{10} + \frac{1}{60}$~~

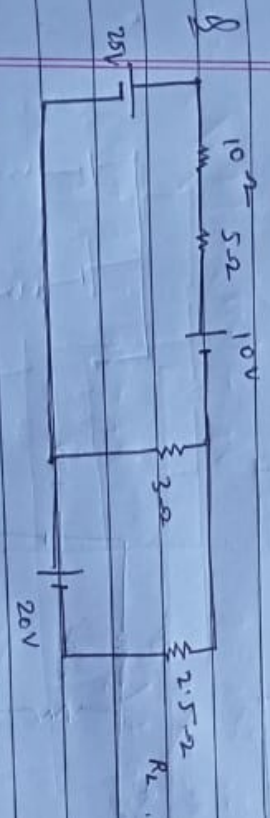
$\frac{1}{6} + \frac{1}{10} + \frac{1}{60} = \frac{10+6+1}{60} = \frac{17}{60}$

R_{th}

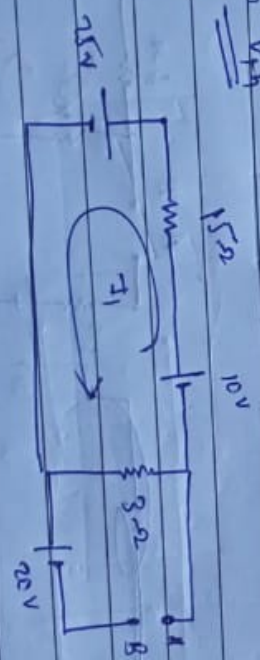


$R_{th} = \frac{140 + 60}{17} = \frac{200}{17} \Omega$

$I_{A-B} = \frac{-110/4}{\frac{200}{17} + 15} = \frac{-110 \times 17}{200 + 105} = \frac{-1870}{305} = -\frac{22}{61} A$



$\rightarrow$ for V_{th}



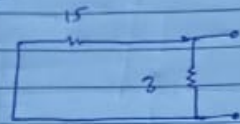
$V_A = \left(\frac{-5}{2} \times 3 \right) - (20)$

$V_A - V_B = 20 \text{ V}$

$25 + 8I_1 - 10 + 15I_1 = 0$
 $18I_1 + 15 = 0$
 $I_1 = -\frac{5}{6} A$

$I_1 = -\frac{5}{6} A$

R_{th}



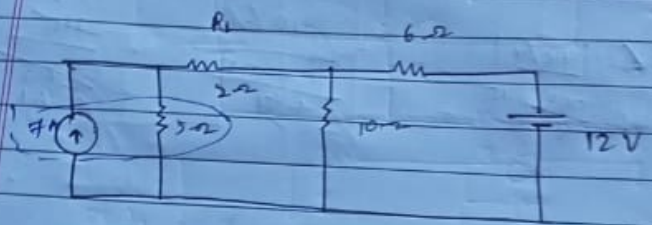
$$\frac{1}{R_{th}} = \frac{1}{15} + \frac{1}{3}$$

$$\frac{1+5}{15} = \frac{6}{15}$$

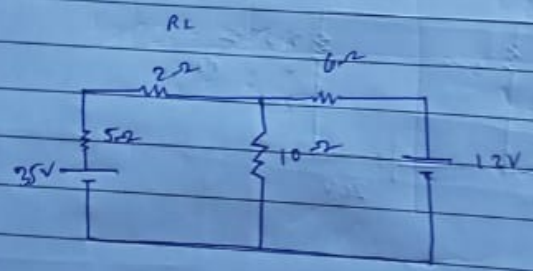
$$R_{th} = \frac{15}{6} = \frac{5}{2}$$

$$I_{th} = \frac{3\sqrt{2}}{\frac{5}{2} + \frac{5}{2}} = \frac{\frac{3\sqrt{2}}{2}}{\frac{10}{2}} = \frac{3\sqrt{2}}{10}$$

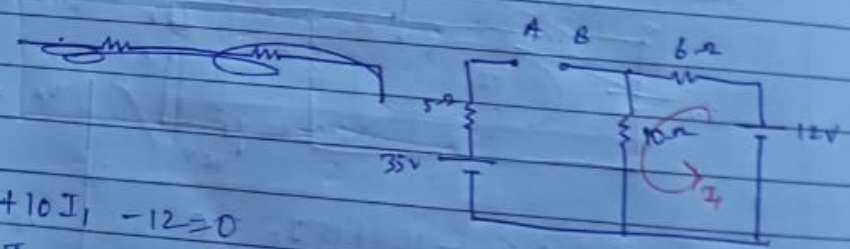
g



$\rightarrow V_{th}$



V_{th}



$$6I_1 + 10I_1 - 12 = 0$$

$$16I_1 = 12$$

$$I_1 = \frac{3}{4}$$

$$\rightarrow V_A - (35) - \left(-\frac{10 \times 10}{2}\right) - V_B = 0$$

$$V_A - V_B = 35 - \frac{15}{2}$$

$$= \frac{70 - 15}{2} = \frac{55}{2} = 27.5 \text{ V}$$

Outer loop

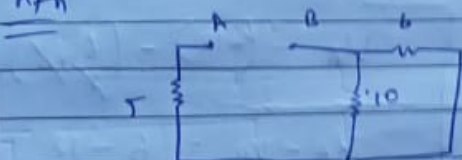
$$V_A - (35) - (12) - \left(\frac{3 \times 3}{2}\right) - V_B = 0$$

$$V_A - V_B = 35 - 12 + \frac{9}{2}$$

$$= 23 + \frac{9}{2} = \frac{46 + 9}{2}$$

$$= \frac{55}{2} = 27.5$$

R_{th}

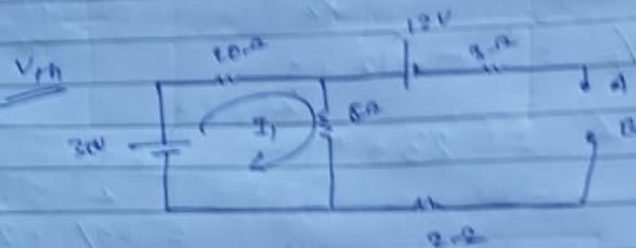
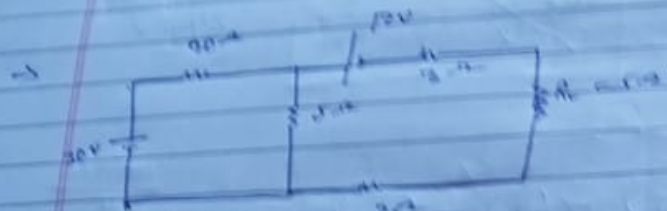
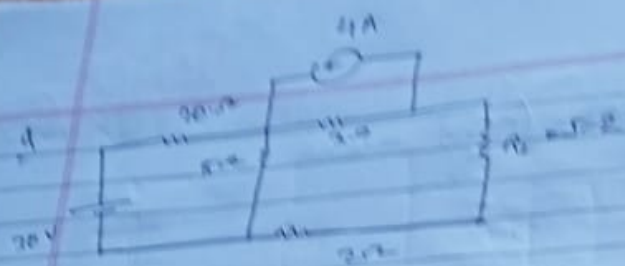


$$\frac{1}{10} + \frac{1}{6} = \frac{6 + 10}{60} = \frac{16}{60}$$

$$R_{eq} = \frac{60 \times 60}{16} = \frac{15}{4}$$

$$R_{th} = \frac{15}{4} + 5 = \frac{35}{4}$$

$$I_{th} = \frac{\frac{55}{2}}{\frac{35}{4} + 2} = \frac{\frac{55}{2}}{\frac{43}{2}} = \frac{55 \times 2}{43} = \frac{110}{43} \text{ A}$$



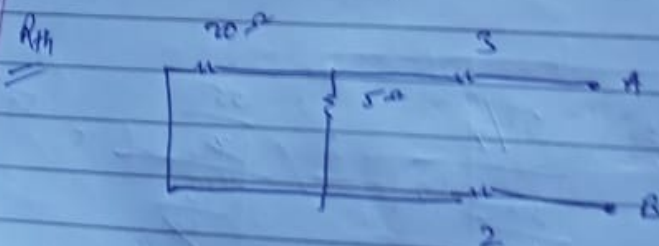
$$20I_1 + 5I_1 - 30 = 0$$

$$5 \cdot 5I_1 = 30 \Rightarrow 6$$

$$I_1 = 6/5$$

$$V_A - (-12) - \left(\frac{6}{5} \cdot 10\right) = 0$$

$$V_A - V_B = 6 - 12 = -6$$



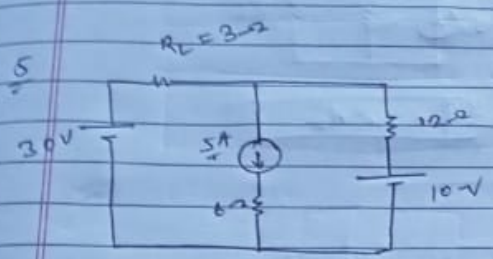
$$\frac{1}{20} + \frac{1}{5}$$

$$\frac{5 + 20}{100} = \frac{25}{100} = \frac{1}{4}$$

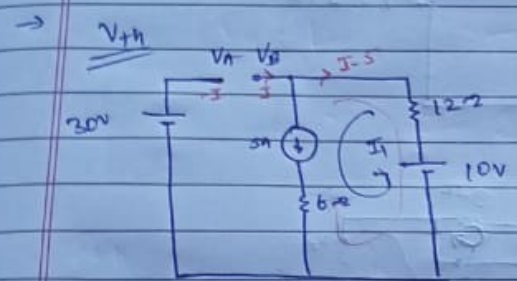
$$R_{eq} = \frac{100}{25} = 4\Omega$$

$$R_{th} = 9\Omega$$

$$I_{ab} = \frac{6-3}{+4-7} = \frac{3}{-7}$$



Since loop can't be formed because C.S. connected to short
 $R = \infty$
 Can't allow how current spread



$$\begin{aligned} 12I_1 + 6(I_1 + 5) - 10 &= 0 \\ 18I_1 - 5 &= 0 \\ I_1 &= 5/18 \end{aligned}$$

$$V_A = (30) \quad 12(I-5) + 10$$

$$I = 50/12$$

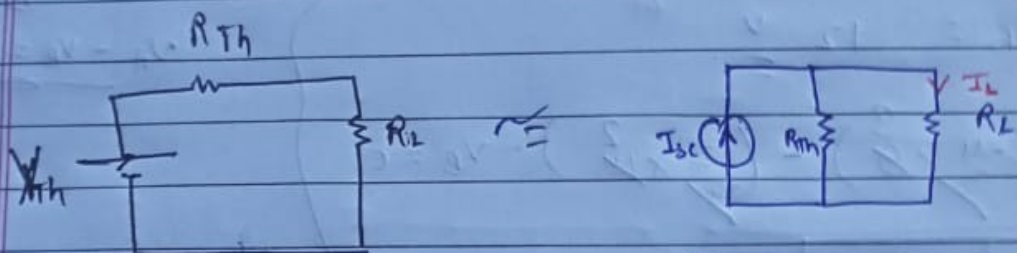
$$I-5 = -5/6$$

$$V_A - (30) - (-10) - (12 \times 5/6) - V_B = 0$$

$$V_A - V_B = 30V$$

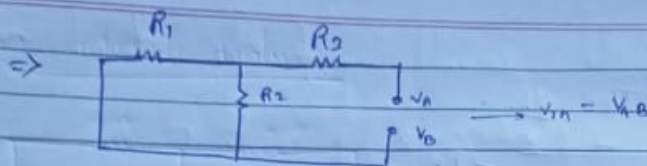
→ Norton Theorem :-

→ Total current = S.C current.



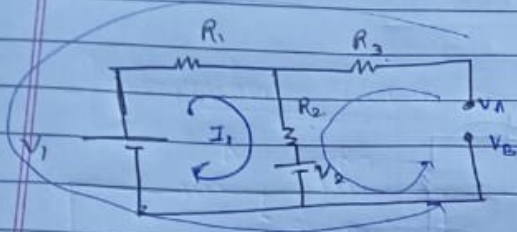
$$R_{Th} = R_N$$

$$I_L = \left(\frac{R_{Th}}{R_{Th} + R_L} \right) I_{sc}$$



$V_{in} \rightarrow R_1 \rightarrow R_{removed}$

Voltage across removed R_2 determine.



$I_1 R_1 + I_1 R_2 + V_2 - V_1 = 0$

$I_1 = \frac{V_1 - V_2}{R_1 + R_2} = A \downarrow$

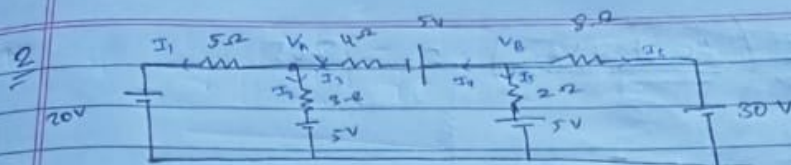
$V_A - (A \cdot R_2) - (V_2) - V_B = 0$

$V_A - V_B = A R_2 + V_2$

Outer Loop

$V_A - (-I_1 R_1) - (V_1) - V_B = 0$

$V_A - V_B = V_1 - I_1 R_1$



At node A
KCL $I_1 + I_2 + I_3 = 0$

$$\frac{V_A - 20}{5} + \frac{V_A - 5}{4} + \frac{V_A - (-5) - V_B}{4} = 0$$

$$\frac{12V_A - 240 + 20V_A - 100 + 15V_A + 75 - 15V_B}{60} = 0$$

$$\frac{47V_A - 340 + 75 - 15V_B}{60} = 0$$

$$\Rightarrow \frac{47V_A - 15V_B - 265}{60} = 0$$

$$\Rightarrow 47V_A - 15V_B = 265$$

$\rightarrow B$

$$\frac{16V_B - 80 - 16V_A + 32V_B - 100 + 8V_B - 240}{64} = 0$$

$$\frac{56V_B - 16V_A - 480}{64} = 0$$

$$7V_B - 2V_A = 60$$

$$\Rightarrow V_B = \frac{60 + 2V_A}{7}$$

$$47V_A - 15V_B = 265$$